

MIT ACADEMY OF ENGINEERING

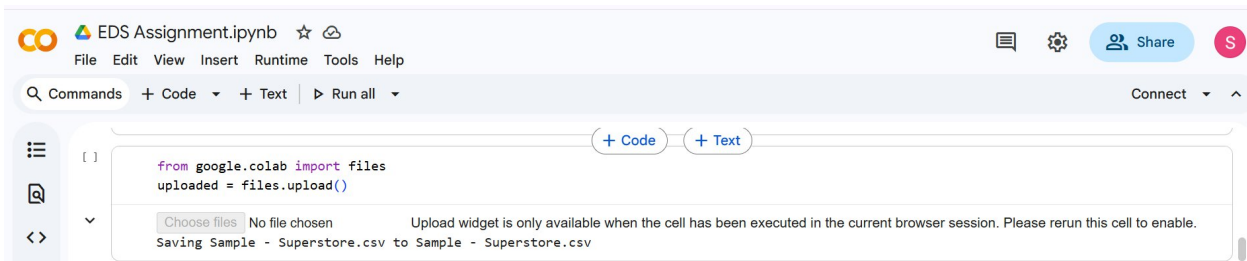
EDS ASSIGNMENT 2

Sales Dataset Analysis

Formulate 20 problem statements for a given dataset using Numpy and Pandas and Apply NumPy and pandas methods to find the solution for the formulated problem statements.

Dataset (Sales Dataset)

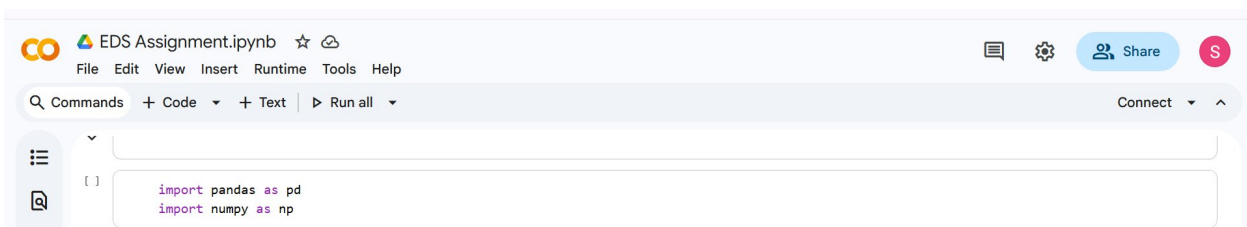
Here Sales data has been successfully downloaded from Kaggle and then uploaded on colab/.



The screenshot shows the Google Colab interface for a notebook titled "EDS Assignment.ipynb". The top bar includes the Colab logo, the notebook name, and icons for chat, settings, share, and a session status indicator (S). Below the top bar is a menu bar with "File", "Edit", "View", "Insert", "Runtime", "Tools", and "Help". A secondary bar contains a search icon, "Commands", and buttons for "+ Code", "+ Text", and "Run all". On the right of this bar is a "Connect" button and a caret icon. The left sidebar shows a file explorer icon, a notebook icon, and a code editor icon. The main area displays a code cell with the following code:

```
from google.colab import files
uploaded = files.upload()
```

 Below the code, a file upload widget is visible, showing "Choose files" and "No file chosen". A message states: "Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable." Below the message, it says "Saving Sample - Superstore.csv to Sample - Superstore.csv". Above the code cell, there are buttons for "+ Code" and "+ Text".



The screenshot shows the Google Colab interface for the same notebook. The top bar and menu bar are identical to the previous screenshot. The left sidebar shows the file explorer icon, a notebook icon, and a code editor icon. The main area displays a code cell with the following code:

```
import pandas as pd
import numpy as np
```

Libraries are imported here.

EDS Assignment.ipynb - Colab x Sales Dataset Analysis x + Ask Gemini

colab.research.google.com/drive/1M_N36i4NhYmJp9qU75fEUHVNY_ICS_v

EDS Assignment.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

```
df = pd.read_csv('Sample - Superstore.csv', encoding='latin1')
df.head()
```

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount	Profit
0	1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gule	Consumer	United States	Henderson	42420	South	FUR-BO-10001798	Furniture	Bookcases	Bush Somerset Collection Bookcase	261.9600	2	0.00	41.9136
1	2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gule	Consumer	United States	Henderson	42420	South	FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs...	731.9400	3	0.00	219.5820
2	3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	90036	West	OFF-LA-10000240	Office Supplies	Labels	Self-Adhesive Address Labels for Typewriters b...	14.6200	2	0.00	6.8714
3	4	US-2015-106966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	33311	South	FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series Slim Rectangular Table	957.5775	5	0.45	-383.0310
4	5	US-2015-106966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	33311	South	OFF-ST-10000760	Office Supplies	Storage	Eldon Fold N Roll Cart	22.3680	2	0.20	2.5164

Variables Terminal

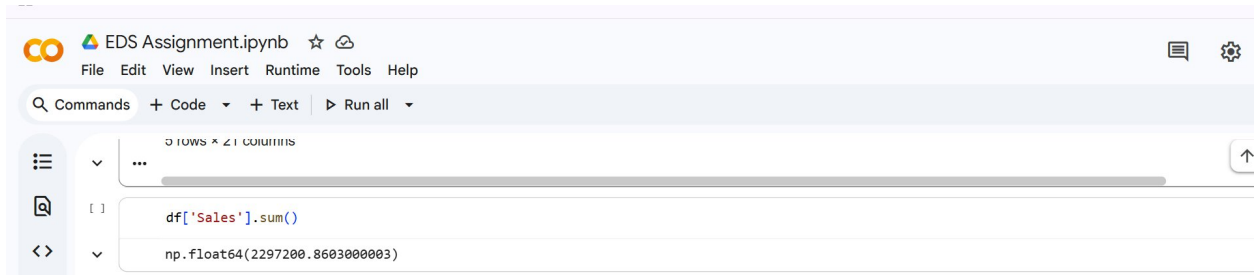
Sales Dataset can be seen having 5 rows × 21 columns.

20 Problem Statements + Solutions

A.Pandas-based (10 Problems)

1. Total sales of the dataset

`df['Sales'].sum()`



EDS Assignment.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

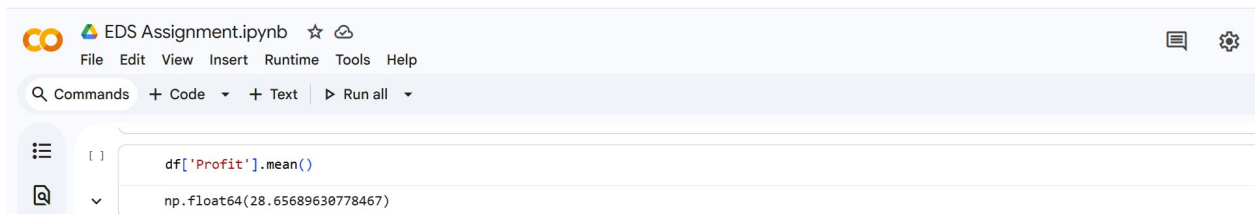
1 ROWS x 21 COLUMNS

```
[ ] df['Sales'].sum()
```

```
[ ] np.float64(2297200.8603000003)
```

2. Average profit

`df['Profit'].mean()`



EDS Assignment.ipynb

File Edit View Insert Runtime Tools Help

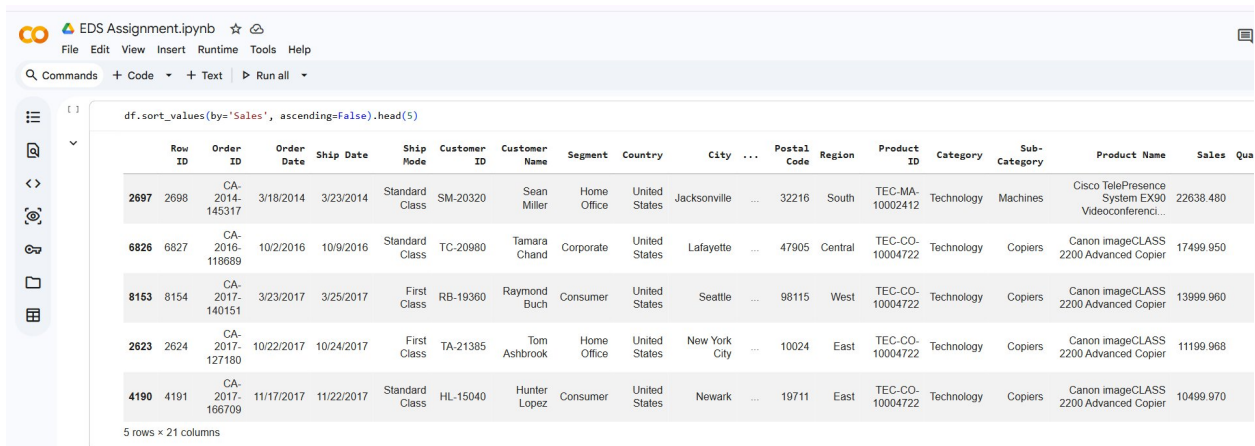
Commands + Code + Text Run all

```
[ ] df['Profit'].mean()
```

```
[ ] np.float64(28.65689630778467)
```

3. Top 5 highest sales records

`df.sort_values(by='Sales', ascending=False).head(5)`



EDS Assignment.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

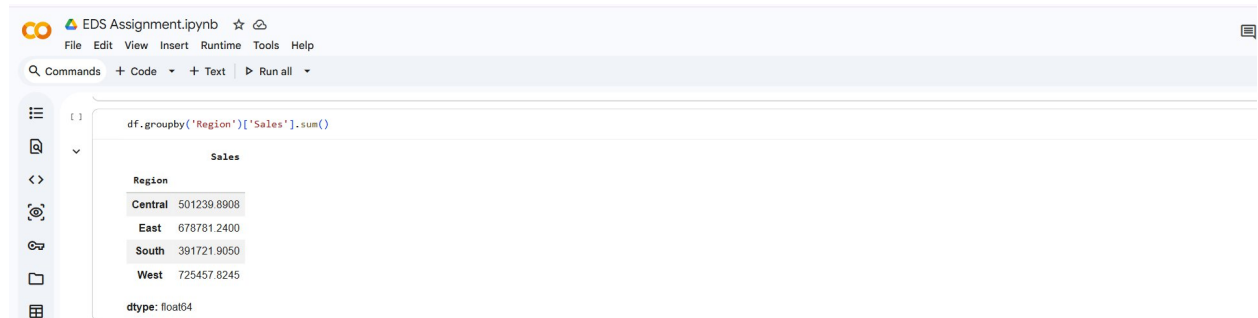
```
[ ] df.sort_values(by='Sales', ascending=False).head(5)
```

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales	Qua
2697	2698	CA-2014-145317	3/18/2014	3/23/2014	Standard Class	SM-20320	Sean Miller	Home Office	United States	Jacksonville	32216	South	TEC-MA-10002412	Technology	Machines	Cisco TelePresence System EX90 Videoconferenci...	22638.480
6826	6827	CA-2016-118689	10/2/2016	10/9/2016	Standard Class	TC-20980	Tamara Chand	Corporate	United States	Lafayette	47905	Central	TEC-CO-10004722	Technology	Copiers	Canon imageCLASS 2200 Advanced Copier	17499.950
8153	8154	CA-2017-140151	3/23/2017	3/25/2017	First Class	RB-19360	Raymond Buch	Consumer	United States	Seattle	98115	West	TEC-CO-10004722	Technology	Copiers	Canon imageCLASS 2200 Advanced Copier	13999.960
2623	2624	CA-2017-127180	10/22/2017	10/24/2017	First Class	TA-21385	Tom Ashbrook	Home Office	United States	New York City	10024	East	TEC-CO-10004722	Technology	Copiers	Canon imageCLASS 2200 Advanced Copier	11199.968
4190	4191	CA-2017-166709	11/17/2017	11/22/2017	Standard Class	HL-15040	Hunter Lopez	Consumer	United States	Newark	10711	East	TEC-CO-10004722	Technology	Copiers	Canon imageCLASS 2200 Advanced Copier	10499.970

5 rows x 21 columns

4. Total sales by region

`df.groupby('Region')['Sales'].sum()`



The screenshot shows a Jupyter Notebook interface with the following code and output:

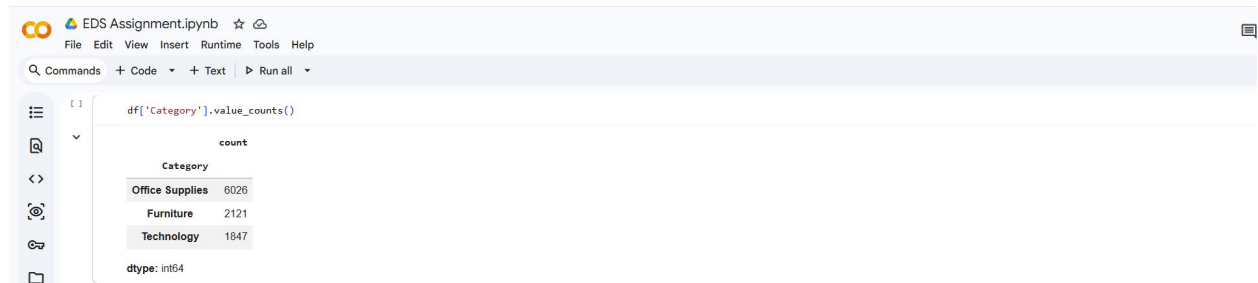
```
df.groupby('Region')['Sales'].sum()
```

Region	Sales
Central	501239.8908
East	678781.2400
South	391721.9050
West	725457.8245

The output is a Series with dtype: float64.

5. Count of orders per category

`df['Category'].value_counts()`



The screenshot shows a Jupyter Notebook interface with the following code and output:

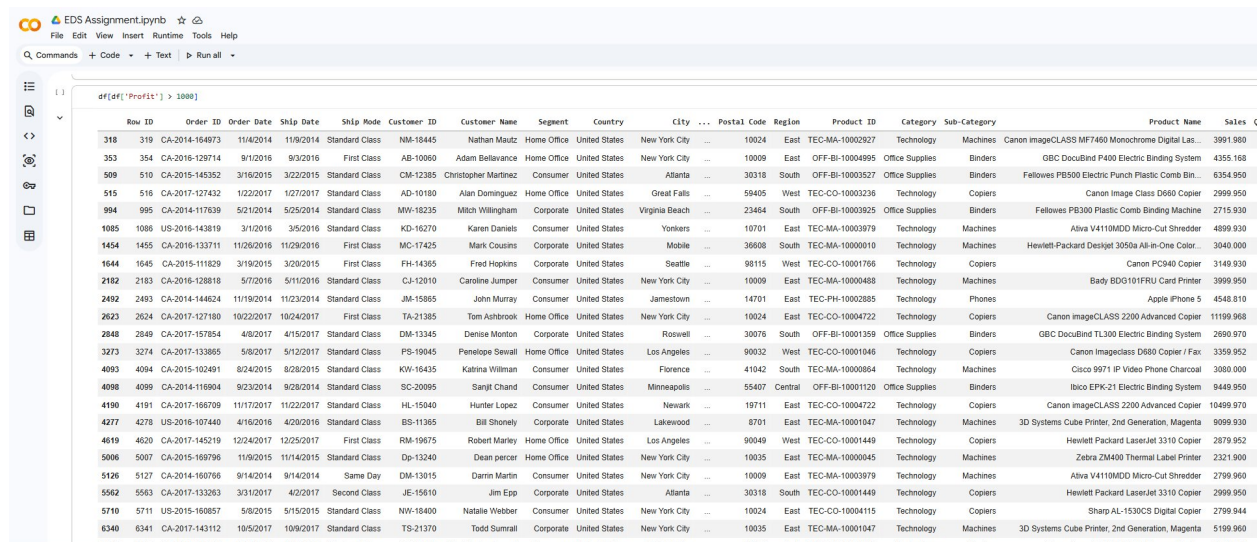
```
df['Category'].value_counts()
```

Category	count
Office Supplies	6026
Furniture	2121
Technology	1847

The output is a Series with dtype: int64.

6. Filter orders with profit > 1000

`df[df['Profit'] > 1000]`



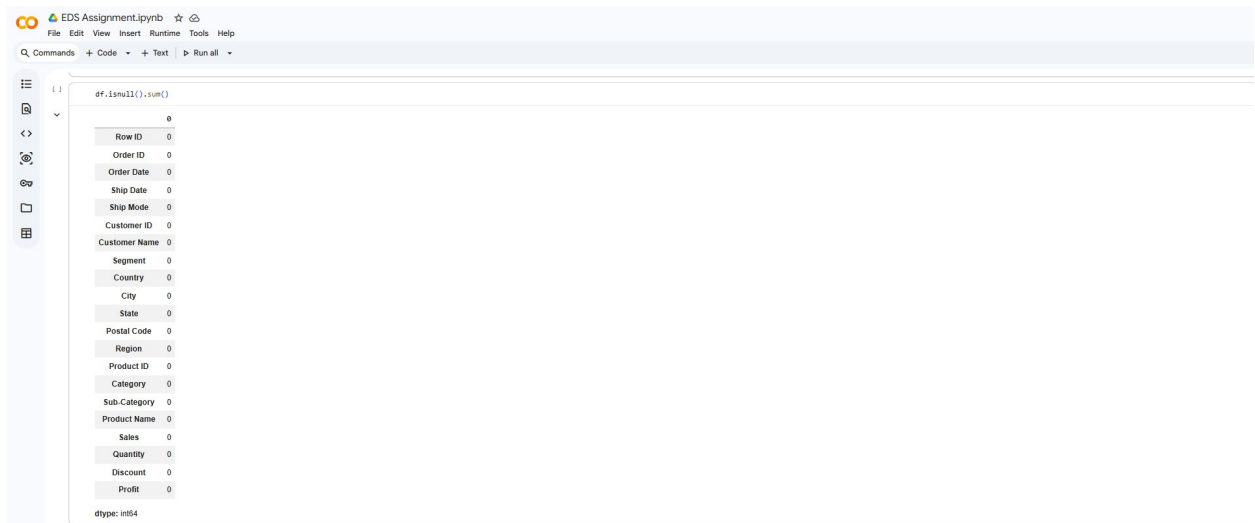
The screenshot shows a Jupyter Notebook interface with the following code and output:

```
df[df['Profit'] > 1000]
```

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales
318	319	CA-2014-164973	11/4/2014	Standard Class	NM-18445	Nathan Mautz	Home Office	United States	New York City	10024	East	TEC-MA-10002927	Technology	Machines	Canon imageCLASS MF7460 Monochrome Digital Las...	3991.980
353	354	CA-2016-129714	9/1/2016	First Class	AB-10060	Adam Bellavance	Home Office	United States	New York City	10099	East	OFF-BI-10004995	Office Supplies	Binders	GBC DocuBind P400 Electric Binding System	4355.168
509	510	CA-2015-145352	3/16/2015	Standard Class	CM-12385	Christopher Martinez	Consumer	United States	Atlanta	30318	South	OFF-BI-10003527	Office Supplies	Binders	Fellowes PB5500 Electric Punch Plastic Comb Bin...	6354.950
515	516	CA-2016-127432	1/22/2017	Standard Class	AD-10180	Alan Dominguez	Home Office	United States	Great Falls	59405	West	TEC-CO-10003236	Technology	Copiers	Canon Image Class D660 Copier	2999.950
994	995	CA-2014-117639	5/21/2014	Standard Class	MW-18235	Mitch Williamson	Corporate	United States	Virginia Beach	23464	South	OFF-BI-10003925	Office Supplies	Binders	Fellowes PB300 Plastic Comb Binding Machine	2715.930
1085	1086	US-2016-143819	3/1/2016	Standard Class	KD-16270	Karen Daniels	Consumer	United States	Yonkers	10701	East	TEC-MA-10003979	Technology	Machines	Altra V4110MDO Micro-Cut Shredder	4899.930
1454	1455	CA-2016-133711	11/26/2016	First Class	MC-17425	Mark Coustons	Corporate	United States	Mobile	36608	South	TEC-MA-10000010	Technology	Machines	Hewlett-Packard DesignJet 3050a All-in-One Color...	3040.000
1644	1645	CA-2015-111829	3/19/2015	First Class	FH-14365	Fred Hopkins	Corporate	United States	Seattle	98115	West	TEC-CO-10001766	Technology	Copiers	Canon PC940 Copier	3149.930
2182	2183	CA-2016-128818	5/7/2016	Standard Class	CJ-12010	Caroline Jumper	Consumer	United States	New York City	10009	East	TEC-MA-10000488	Technology	Machines	Bady EDG101FRU Card Printer	3999.950
2492	2493	CA-2014-144624	11/19/2014	Standard Class	JM-15965	John Murray	Consumer	United States	Jamestown	14701	East	TEC-PH-10002885	Technology	Phones	Apple iPhone 5	4548.810
2623	2624	CA-2017-127180	10/22/2017	First Class	TA-21385	Tom Ashbrook	Home Office	United States	New York City	10024	East	TEC-CO-10004722	Technology	Copiers	Canon imageCLASS 2200 Advanced Copier	11199.968
2848	2849	CA-2017-157854	4/6/2017	Standard Class	DM-13345	Denise Monton	Corporate	United States	Roswell	30076	South	OFF-BI-10001359	Office Supplies	Binders	GBC DocuBind TL300 Electric Binding System	2690.970
3273	3274	CA-2017-133855	5/9/2017	Standard Class	PS-19045	Penelope Sewall	Home Office	United States	Los Angeles	90032	West	TEC-CO-10001046	Technology	Copiers	Canon imageCLASS D680 Copier / Fax	3359.952
4093	4094	CA-2015-102491	8/24/2015	Standard Class	KW-16435	Katrina Wilman	Consumer	United States	Florence	41042	South	TEC-MA-10000864	Technology	Machines	Cisco 9971 IP Video Phone Charcoal	3080.000
4098	4099	CA-2014-116904	9/23/2014	Standard Class	SC-20095	Sanjit Chand	Consumer	United States	Minneapolis	55407	Central	OFF-BI-10001129	Office Supplies	Binders	Ibico EPK-21 Electric Binding System	9449.950
4190	4191	CA-2017-166709	11/17/2017	Standard Class	HL-15040	Hunter Lopez	Consumer	United States	Newark	19711	East	TEC-CO-10004722	Technology	Copiers	Canon imageCLASS 2200 Advanced Copier	10499.970
4277	4278	US-2016-107440	4/19/2016	Standard Class	BS-11365	Bill Shonely	Corporate	United States	Lakewood	8701	East	TEC-MA-10001047	Technology	Machines	3D Systems Cube Printer, 2nd Generation, Magenta	9099.930
4619	4620	CA-2014-145219	12/24/2017	First Class	RM-19675	Robert Marley	Home Office	United States	Los Angeles	90049	West	TEC-CO-10001449	Technology	Copiers	Hewlett-Packard LaserJet 3310 Copier	2879.952
5006	5007	CA-2015-169796	11/9/2015	Standard Class	Dp-13240	Dean percer	Home Office	United States	New York City	10035	East	TEC-MA-10000045	Technology	Machines	Zebra ZM400 Thermal Label Printer	2321.900
5126	5127	CA-2014-160766	9/14/2014	Same Day	DM-13015	Darrin Martin	Consumer	United States	New York City	10009	East	TEC-MA-10003979	Technology	Machines	Altra V4110MDO Micro-Cut Shredder	2799.960
5562	5563	CA-2017-133263	3/31/2017	Second Class	JE-15610	Jim Epp	Corporate	United States	Atlanta	30318	South	TEC-CO-10001449	Technology	Copiers	Hewlett-Packard LaserJet 3310 Copier	2999.950
5710	5711	US-2015-160857	5/6/2015	Standard Class	NW-18400	Natalie Webber	Consumer	United States	New York City	10024	East	TEC-CO-10004115	Technology	Copiers	Sharp AL-1530CS Digital Copier	2799.944
6340	6341	CA-2017-131112	10/5/2017	Standard Class	TS-21370	Todd Sumrall	Corporate	United States	New York City	10035	East	TEC-MA-10001047	Technology	Machines	3D Systems Cube Printer, 2nd Generation, Magenta	9199.960
6436	6437	CA-2016-143714	6/23/2016	Standard Class	CP-13370	Christopher Carroll	Consumer	United States	Birmingham	35210	East	TEC-CO-10004723	Technology	Copiers	Canon imageCLASS 2200 Advanced Copier	9300.036

7. Find null values in dataset

df.isnull().sum()



EDS Assignment.ipynb

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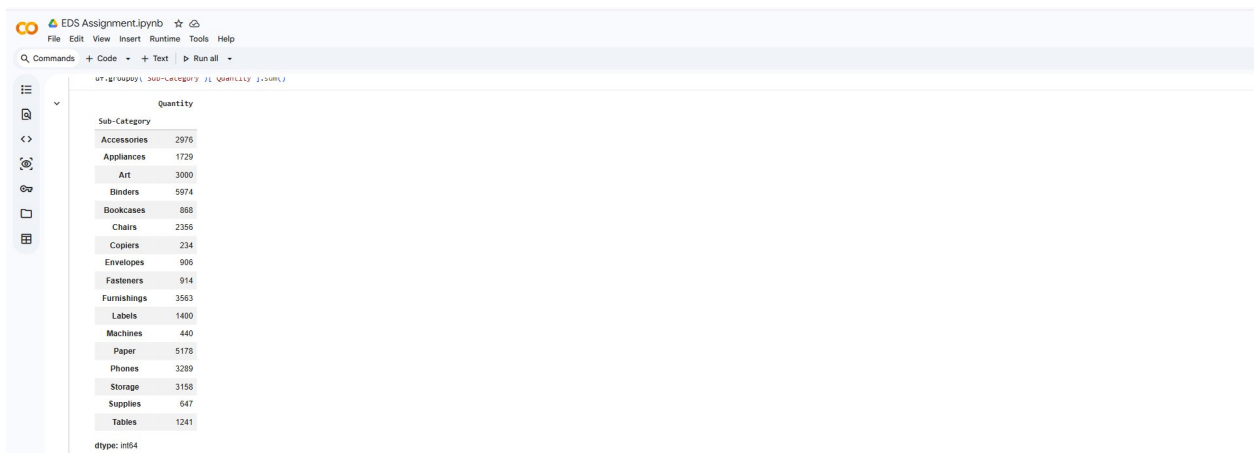
```
df.isnull().sum()
```

	0
Row ID	0
Order ID	0
Order Date	0
Ship Date	0
Ship Mode	0
Customer ID	0
Customer Name	0
Segment	0
Country	0
City	0
State	0
Postal Code	0
Region	0
Product ID	0
Category	0
Sub-Category	0
Product Name	0
Sales	0
Quantity	0
Discount	0
Profit	0

dtype: int64

8. Total quantity sold per sub-category

df.groupby('Sub-Category')['Quantity'].sum()



EDS Assignment.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

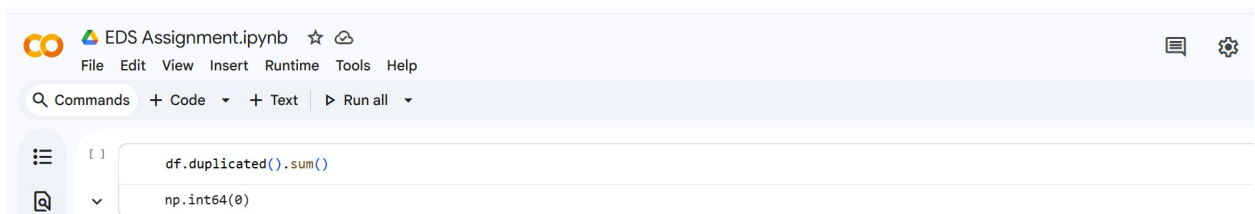
```
df.groupby('Sub-Category')['Quantity'].sum()
```

Sub-Category	Quantity
Accessories	2976
Appliances	1729
Art	3000
Binders	5974
Bookcases	868
Chairs	2356
Copiers	234
Envelopes	906
Fasteners	914
Furnishings	3563
Labels	1400
Machines	440
Paper	5178
Phones	3288
Storage	3158
Supplies	647
Tables	1241

dtype: int64

9. Find duplicate rows

df.duplicated().sum()



EDS Assignment.ipynb

File Edit View Insert Runtime Tools Help

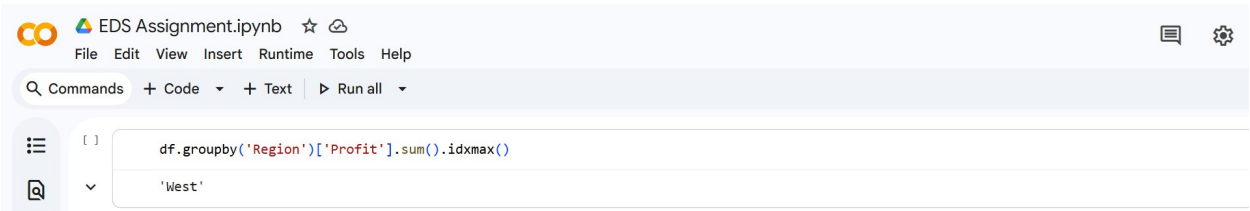
Commands + Code + Text Run all

```
df.duplicated().sum()
```

np.int64(0)

10. Region with highest profit

`df.groupby('Region')['Profit'].sum().idxmax()`



The screenshot shows a Jupyter Notebook interface. At the top, the title bar reads "EDS Assignment.ipynb" with a star icon and a refresh icon. Below the title bar is a menu bar with "File", "Edit", "View", "Insert", "Runtime", "Tools", and "Help". A toolbar below the menu bar contains a search icon, "Commands", "+ Code", "+ Text", and "Run all". The notebook area shows a code cell with the command `df.groupby('Region')['Profit'].sum().idxmax()`. To the left of the code cell is a sidebar with a list icon and a search icon. Below the code cell, the output is displayed as `'West'`.

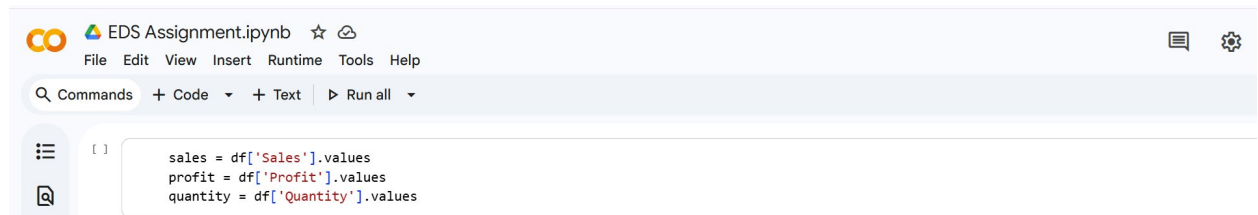
B. NumPy-based (10 Problems)

(Convert columns into arrays)

```
sales = df['Sales'].values
```

```
profit = df['Profit'].values
```

```
quantity = df['Quantity'].values
```

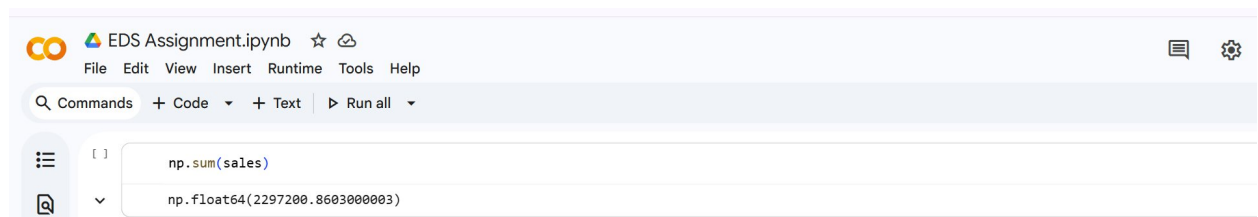


The screenshot shows a Jupyter Notebook titled "EDS Assignment.ipynb". The code cell contains the following lines:

```
sales = df['Sales'].values
profit = df['Profit'].values
quantity = df['Quantity'].values
```

11. Total sales using NumPy

```
np.sum(sales)
```



The screenshot shows the Jupyter Notebook interface with the code cell containing:

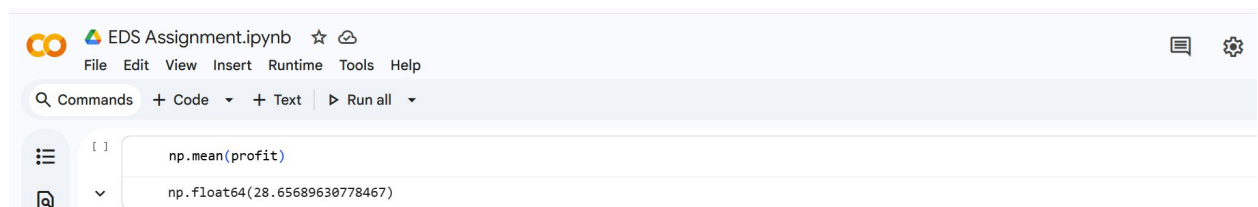
```
np.sum(sales)
```

The output cell displays the result:

```
np.float64(2297200.8603000003)
```

12. Mean profit

```
np.mean(profit)
```



The screenshot shows the Jupyter Notebook interface with the code cell containing:

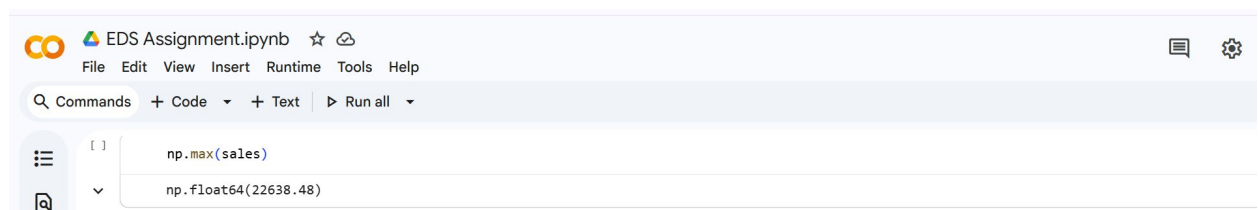
```
np.mean(profit)
```

The output cell displays the result:

```
np.float64(28.65689630778467)
```

13. Maximum sales value

```
np.max(sales)
```



The screenshot shows the Jupyter Notebook interface with the code cell containing:

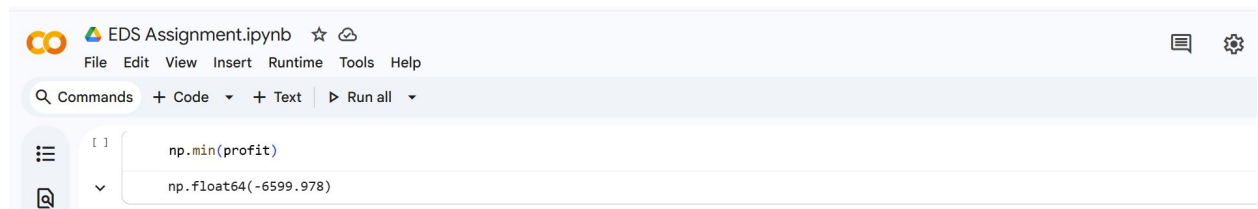
```
np.max(sales)
```

The output cell displays the result:

```
np.float64(22638.48)
```

14. Minimum profit

np.min(profit)

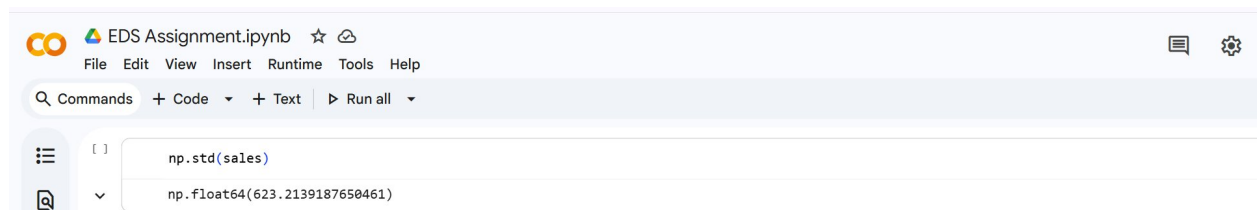


The image shows a Jupyter Notebook interface with the title "EDS Assignment.ipynb". The menu bar includes "File", "Edit", "View", "Insert", "Runtime", "Tools", and "Help". Below the menu bar is a toolbar with "Commands", "+ Code", "+ Text", and "Run all". The notebook contains a single code cell with the following code:

```
[ ] np.min(profit)
np.float64(-6599.978)
```

15. Standard deviation of sales

np.std(sales)

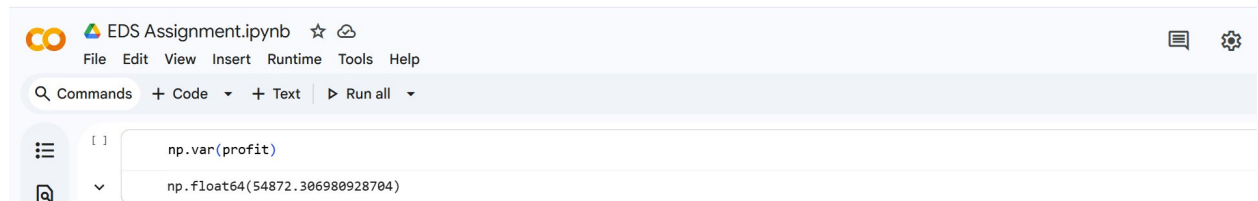


The image shows a Jupyter Notebook interface with the title "EDS Assignment.ipynb". The menu bar includes "File", "Edit", "View", "Insert", "Runtime", "Tools", and "Help". Below the menu bar is a toolbar with "Commands", "+ Code", "+ Text", and "Run all". The notebook contains a single code cell with the following code:

```
[ ] np.std(sales)
np.float64(623.2139187650461)
```

16. Variance of profit

np.var(profit)

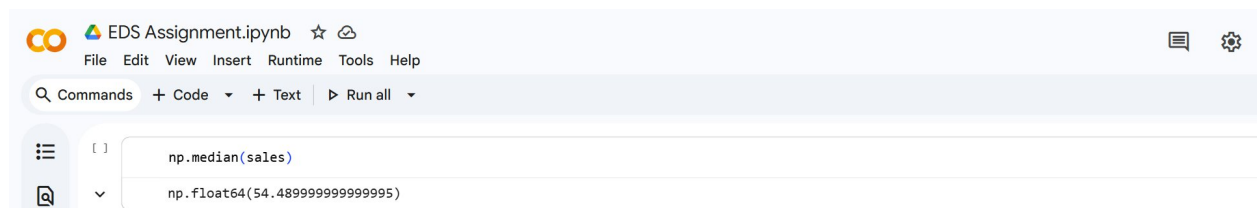


The image shows a Jupyter Notebook interface with the title "EDS Assignment.ipynb". The menu bar includes "File", "Edit", "View", "Insert", "Runtime", "Tools", and "Help". Below the menu bar is a toolbar with "Commands", "+ Code", "+ Text", and "Run all". The notebook contains a single code cell with the following code:

```
[ ] np.var(profit)
np.float64(54872.306980928704)
```

17. Find median sales

np.median(sales)

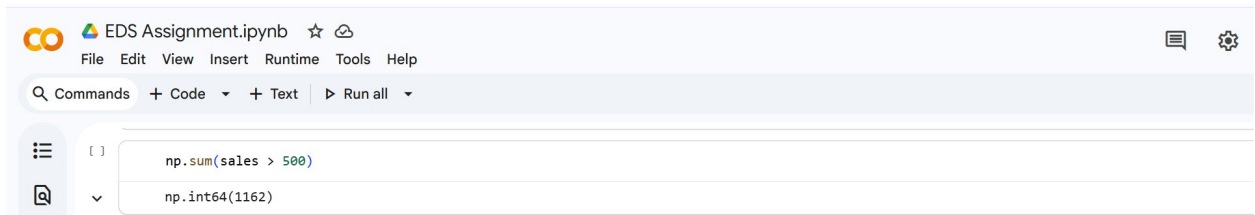


The image shows a Jupyter Notebook interface with the title "EDS Assignment.ipynb". The menu bar includes "File", "Edit", "View", "Insert", "Runtime", "Tools", and "Help". Below the menu bar is a toolbar with "Commands", "+ Code", "+ Text", and "Run all". The notebook contains a single code cell with the following code:

```
[ ] np.median(sales)
np.float64(54.489999999999995)
```

18. Count of values > 500 sales

`np.sum(sales > 500)`

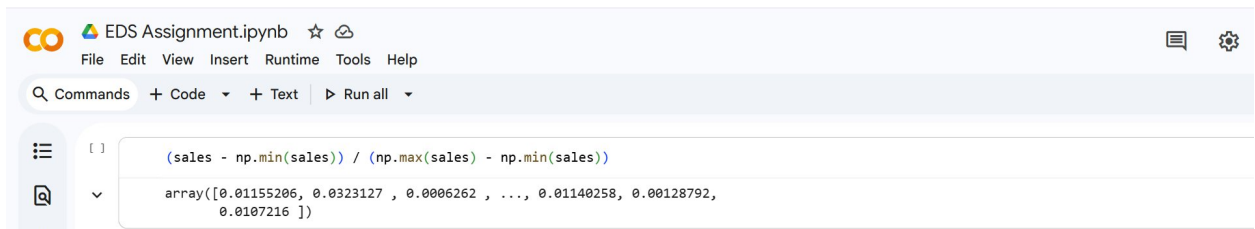


The image shows a Jupyter Notebook interface with the title "EDS Assignment.ipynb". The menu bar includes File, Edit, View, Insert, Runtime, Tools, and Help. The toolbar shows a search icon, "Commands", and buttons for "+ Code", "+ Text", and "Run all". The code cell contains the following code:

```
[ ] np.sum(sales > 500)
np.int64(1162)
```

19. Normalize sales data

`(sales - np.min(sales)) / (np.max(sales) - np.min(sales))`

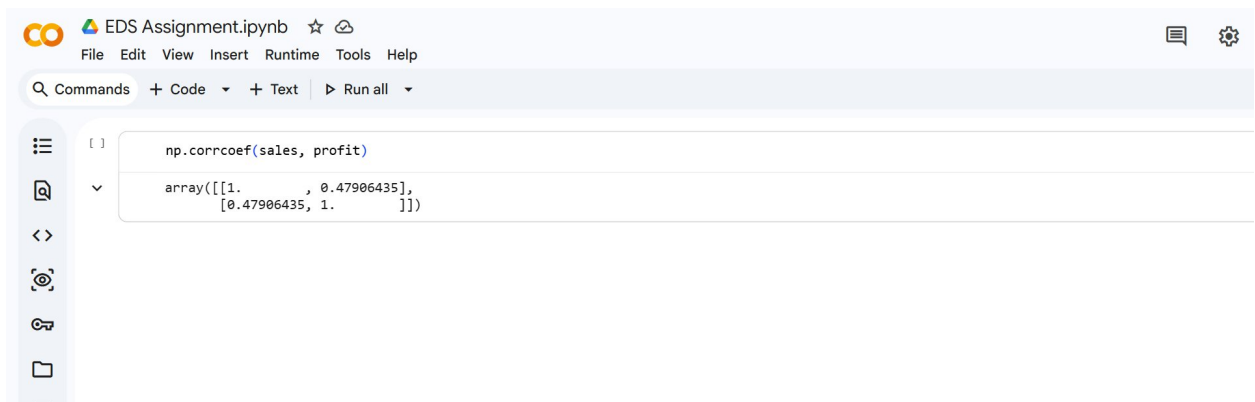


The image shows a Jupyter Notebook interface with the title "EDS Assignment.ipynb". The menu bar includes File, Edit, View, Insert, Runtime, Tools, and Help. The toolbar shows a search icon, "Commands", and buttons for "+ Code", "+ Text", and "Run all". The code cell contains the following code:

```
[ ] (sales - np.min(sales)) / (np.max(sales) - np.min(sales))
array([0.01155206, 0.0323127 , 0.0006262 , ..., 0.01140258, 0.00128792,
       0.0107216 ])
```

20. Correlation between sales and profit

`np.corrcoef(sales, profit)`



The image shows a Jupyter Notebook interface with the title "EDS Assignment.ipynb". The menu bar includes File, Edit, View, Insert, Runtime, Tools, and Help. The toolbar shows a search icon, "Commands", and buttons for "+ Code", "+ Text", and "Run all". The code cell contains the following code:

```
[ ] np.corrcoef(sales, profit)
array([[1.          , 0.47906435],
       [0.47906435, 1.          ]])
```